

BedLog Raspberry Pi Installation Guide

Backend + PostgreSQL database + Vite frontend + Chromium kiosk + automatic landscape rotation

Target platform: Raspberry Pi running Debian 13 (Trixie), LightDM, Raspberry Pi Desktop rpd-labwc / Wayland, DSI display and user pi.

This guide consolidates the installation steps verified on BedLog2. The target boot flow is: PostgreSQL -> BedLog backend -> Vite frontend -> LightDM/labwc -> landscape display -> Chromium kiosk.

1. Project layout

The BedLog project is expected at /home/pi/Documents/bedlog. Important files include:

```
/home/pi/Documents/bedlog/  
package.json  
vite.config.ts  
src/  
dist/  
db-backend/  
  package.json  
  server.cjs  
  schema.sql  
esp-bridge/  
  server.cjs
```

2. Install required software

```
sudo apt update  
sudo apt install -y nodejs npm postgresql postgresql-contrib chromium
```

Verify the installation:

```
node -v  
npm -v  
psql --version  
chromium --version  
sudo systemctl status postgresql --no-pager  
pg_lsclusters
```

On the verified BedLog2 installation, Node was /usr/bin/node, npm was /usr/bin/npm, and PostgreSQL 17 used port 5432.

3. Create the PostgreSQL database

Create a local PostgreSQL role and database named bedlog:

```
sudo -u postgres psql
```

At the PostgreSQL prompt:

```
CREATE USER bedlog WITH PASSWORD 'bedlog';  
CREATE DATABASE bedlog OWNER bedlog;  
\q
```

Load the supplied schema:

```
cd /home/pi/Documents/bedlog  
  
PGPASSWORD=bedlog psql \  
-h localhost \  
-U bedlog \  
-d bedlog \  
-f db-backend/schema.sql
```

Verify the tables:

```
PGPASSWORD=bedlog psql -h localhost -U bedlog -d bedlog -c '\dt'
```

If authentication fails, reset the local role password:

```
sudo -u postgres psql  
ALTER USER bedlog WITH PASSWORD 'bedlog';  
\q
```

Then verify:

```
PGPASSWORD=bedlog psql \  
-h localhost -p 5432 -U bedlog -d bedlog \  
-c "SELECT current_user, current_database();"
```

4. Prepare frontend dependencies for Raspberry Pi

If the project was copied from an Apple Silicon Mac, remove explicitly Mac-only packages before reinstalling dependencies.

```
cd /home/pi/Documents/bedlog

npm pkg delete devDependencies.@rollup/rollup-darwin-arm64
npm pkg delete devDependencies.lightningcss-darwin-arm64

rm -rf node_modules
rm -f package-lock.json

npm install
npm run build
```

Verify that the build generated the dist directory:

```
ls -lah /home/pi/Documents/bedlog/dist
```

5. Install and test the database backend

```
cd /home/pi/Documents/bedlog/db-backend
rm -rf node_modules
npm install
```

For a manual test:

```
export PGHOST=localhost
export PGPORT=5432
export PGDATABASE=bedlog
export PGUSER=bedlog
export PGPASSWORD=bedlog
```

```
npm start
```

From another terminal:

```
curl -i http://localhost:4002/health
```

Expected result:

```
HTTP/1.1 200 OK
...
{"ok":true}
```

The current db-backend implements /health, POST /api/backup, and GET /api/backup/.
/api/display/summary is not present in this version and will return 404 until that route is added.

6. Test the Vite frontend

```
cd /home/pi/Documents/bedlog
npm run dev -- --host 0.0.0.0
```

From another terminal:

```
curl -I http://localhost:5173
```

The expected response is HTTP 200.

7. Create the backend boot service

Create /etc/systemd/system/bedlog-backend.service:

```
sudo nano /etc/systemd/system/bedlog-backend.service

[Unit]
Description=BedLog PostgreSQL Backend
After=network.target postgresql.service
Requires=postgresql.service
```

```
[Service]
Type=simple
User=pi
Group=pi
WorkingDirectory=/home/pi/Documents/bedlog/db-backend

Environment=NODE_ENV=production
Environment=PORT=4002
Environment=PGHOST=localhost
Environment=PGPORT=5432
Environment=PGDATABASE=bedlog
Environment=PGUSER=bedlog
Environment=PGPASSWORD=bedlog

ExecStart=/usr/bin/node /home/pi/Documents/bedlog/db-backend/server.cjs

Restart=always
RestartSec=3

[Install]
WantedBy=multi-user.target
```

Enable and start it:

```
sudo systemctl daemon-reload
sudo systemctl enable --now bedlog-backend.service

systemctl status bedlog-backend.service --no-pager
curl -i http://localhost:4002/health
```

8. Create the Vite frontend boot service

Create /etc/systemd/system/bedlog-vite.service:

```
sudo nano /etc/systemd/system/bedlog-vite.service

[Unit]
Description=BedLog Vite Frontend
After=network.target bedlog-backend.service
Wants=network-online.target
Requires=bedlog-backend.service

[Service]
Type=simple
User=pi
Group=pi
WorkingDirectory=/home/pi/Documents/bedlog

Environment=NODE_ENV=development
Environment=HOME=/home/pi

ExecStart=/usr/bin/npm run dev -- --host 0.0.0.0 --port 5173

Restart=always
RestartSec=3

[Install]
WantedBy=multi-user.target
```

Enable and test:

```
sudo systemctl daemon-reload
sudo systemctl enable --now bedlog-vite.service

systemctl status bedlog-vite.service --no-pager
curl -I http://localhost:5173
```

9. Configure automatic landscape rotation

BedLog2 uses a DSI display identified as DSI-1. The verified landscape command is:

```
wlr-randr --output DSI-1 --transform 90
```

Create the labwc autostart file if necessary:

```
mkdir -p /home/pi/.config/labwc
touch /home/pi/.config/labwc/autostart
```

Add a short delay so the DSI output is ready after login:

```
echo '(sleep 2; wlr-randr --output DSI-1 --transform 90) &' \
>> /home/pi/.config/labwc/autostart
```

Verify:

```
cat /home/pi/.config/labwc/autostart
wlr-randr
```

The output should show DSI-1 and Transform: 90.

10. Configure Chromium kiosk

The verified desktop configuration is LightDM autologin to user pi with session rpd-labwc. Chromium therefore needs to join the existing Wayland session.

Create /etc/systemd/system/bedlog-kiosk.service:

```
sudo nano /etc/systemd/system/bedlog-kiosk.service

[Unit]
Description=BedLog Chromium Kiosk
After=graphical.target lightdm.service bedlog-vite.service
Wants=graphical.target
Requires=bedlog-vite.service

[Service]
Type=simple
User=pi
Group=pi

Environment=HOME=/home/pi
Environment=XDG_RUNTIME_DIR=/run/user/1000
Environment=WAYLAND_DISPLAY=wayland-0

ExecStartPre=/bin/bash -c 'for i in {1..30}; do [ -S /run/user/1000/wayland-0 ] && exit 0; sleep 1; done; exit 1'

ExecStart=/usr/bin/chromium \
  --ozone-platform=wayland \
  --user-data-dir=/home/pi/.config/bedlog-chromium \
  --kiosk \
  --no-first-run \
  --disable-session-crashed-bubble \
  --disable-infobars \
  --noerrdialogs \
  http://localhost:5173

Restart=on-failure
RestartSec=5

[Install]
WantedBy=graphical.target
```

Enable and start:

```
sudo systemctl daemon-reload
sudo systemctl enable bedlog-kiosk.service
sudo systemctl start bedlog-kiosk.service

systemctl status bedlog-kiosk.service --no-pager -l
```

11. Verify the complete stack

```
echo "===== ROTATION AUTOSTART ====="
cat /home/pi/.config/labwc/autostart
```

```

echo
echo "===== POSTGRESQL ====="
pg_lsclusters

echo
echo "===== BACKEND ====="
systemctl is-enabled bedlog-backend
systemctl is-active bedlog-backend
curl -s http://localhost:4002/health
echo

echo
echo "===== FRONTEND ====="
systemctl is-enabled bedlog-vite
systemctl is-active bedlog-vite
curl -s -o /dev/null -w "HTTP %{http_code}\n" http://localhost:5173

echo
echo "===== KIOSK ====="
systemctl is-enabled bedlog-kiosk
systemctl is-active bedlog-kiosk

echo
echo "===== PORTS ====="
sudo ss -lntp | grep -E ':5432|:4002|:5173'

echo
echo "===== FAILED SERVICES ====="
systemctl --failed --no-pager

```

12. Full reboot test

When all checks are successful, reboot:

```
sudo reboot
```

Do not manually start any BedLog component. The expected automatic sequence is:

Stage	Expected result
1	PostgreSQL 17 cluster starts on port 5432
2	bedlog-backend.service starts on port 4002
3	bedlog-vite.service starts on port 5173
4	LightDM automatically logs in user pi
5	labwc starts and rotates DSI-1 to Transform 90
6	Chromium starts in kiosk mode
7	BedLog opens at http://localhost:5173 in landscape fullscreen

13. Troubleshooting commands

Backend/database failure

```

sudo systemctl status bedlog-backend.service --no-pager -l
sudo journalctl -u bedlog-backend.service -n 50 --no-pager
PGPASSWORD=bedlog psql -h localhost -U bedlog -d bedlog -c "SELECT 1;"
curl -i http://localhost:4002/health

```

Frontend failure

```

sudo systemctl status bedlog-vite.service --no-pager -l
sudo journalctl -u bedlog-vite.service -n 50 --no-pager
curl -I http://localhost:5173

```

Kiosk failure

```
sudo systemctl status bedlog-kiosk.service --no-pager -l
sudo journalctl -u bedlog-kiosk.service -n 50 --no-pager
ls -la /run/user/1000/wayland-*
ps aux | grep '[c]hromium'
```

Display rotation failure

```
wlr-randr
wlr-randr --output DSI-1 --transform 90
cat /home/pi/.config/labwc/autostart
```

14. Verified BedLog2 baseline

At the point this guide was prepared, the verified baseline was:

```
PostgreSQL:         online / enabled
Backend :4002:       enabled / active
Backend /health:    {"ok":true}
Frontend :5173:      enabled / active / HTTP 200
Chromium kiosk:     enabled / active
Display:            DSI-1
Resolution:         720x1280 native
Landscape transform: 90
Desktop:            LightDM + rpd-labwc / Wayland
Autologin user:     pi
```

Before cloning this Raspberry Pi to additional BedLog devices, complete the application-level verification: persistent BedLog data, Mediboard/cloud synchronization, device-specific data cleanup, and any required ESP32 `/api/display/summary` endpoint.